

Filippo Maggioli, Ph.D.

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Short Bio

📌 Research profile

I am a Tenured Assistant Professor at *Pegaso University*. Previously, I was Postdoctoral Researcher at *University of Milano-Bicocca* in the *DIG AIR* research lab led by Simone Melzi, a Postdoctoral Researcher at *Sapienza – University of Rome* in the *GLADIA* research lab led by Emanuele Rodolà, and a Research Intern at the *King Abdullah University of Science and Technology (KAUST)* in the *VCC* research lab led by Peter Wonka. I received my Ph.D. in Computer Science at *Sapienza – University of Rome* (2023), where I also graduated in Computer Science (2019).

I work on geometry processing, spectral geometry, and 3D shape analysis, but I am an active researcher also in other fields of computer graphics, such as procedural shading and physical simulation.

I regularly serve in the program committee of international conferences as chair and reviewer, and I maintain worldwide collaborations with researchers from other institutions and countries.

📌 Research interests

Geometry Processing; Spectral Geometry; 3D Shape Analysis; Procedural Texturing; Simulation of Natural Phenomena; Numerical Linear Algebra.

📌 Author profiles

ORCID:  0000-0001-8008-8468

Google Scholar ID: VN1fbwUAAAAJ *h*-index: 7 *i*10-index: 5

Scopus Author ID: 57216313662 *h*-index: 5 *i*10-index: 3

Academic Appointments

- | | |
|---------------------|--|
| Mar 2025 – Present | 📌 Assistant professor - tenure track. Pegaso University
Department of Computer Science and Information Technologies. |
| Dec 2023 – Feb 2025 | 📌 Postdoctoral researcher. University of Milano-Bicocca
Member of the <i>DIG AIR</i> research lab.
Advisor: prof. Simone Melzi.
Research activity on computational and spectral geometry. |
| Apr 2024 – Feb 2025 | 📌 Adjunct professor. Pegaso University
Undergrad courses on <i>Computer Architecture</i> and <i>Networking and Cybersecurity</i> .
Supervision of BSc students during the development of their theses. |
| Aug 2023 – Nov 2023 | 📌 Postdoctoral researcher. Sapienza – University of Rome
Member of the <i>Gladia</i> research lab.
Advisor: prof. Emanuele Rodolà.
Research activity on computational geometry, spectral geometry, and numerical linear algebra. |
| Sep 2022 – Jan 2023 | 📌 Research internship. King Abdullah University of Science and Technology
Member of the <i>VCC</i> research lab.
Advisor: prof. Dominik L. Michels.
Research activity on simulation of natural phenomena in agricultural settings. |
| Mar 2021 – Jul 2021 | 📌 Teaching assistant. Sapienza – University of Rome
Undergrad course on <i>Introduction to Algorithms</i> . |

Education

- 2019 – 2023  **Ph.D. in Computer Science.** Sapienza – University of Rome.
Advisor: prof. Emanuele Rodolà.
Thesis title: *Scalable geometry processing for computer graphics applications.*
Honourable mention at EG-Italy Award for PhD Thesis in Computer Graphics.
- 2018 – 2019  **M.Sc. in Computer Science.** Sapienza – University of Rome.
Advisor: prof. Emanuele Rodolà.
Thesis title: *Time-efficient function reconstruction via Laplacian eigenproducts.*
- 2014 – 2017  **B.Sc. in Computer Science.** Sapienza – University of Rome.
Advisor: prof. Enrico Tronci.
Thesis title: *Modeling of biological pathways with systems of differential-algebraic equations.*

Courses and Schools

- Jul 2022  **IRDTA DeepLearn 2022 Summer.**
6th International Gran Canaria School on Deep Learning. *Las Palmas de Gran Canaria, Spain.*
- Jul 2021  **ACDL 2021.**
Advanced Online & Onsite Course on Data Science & Machine Learning. *Pontignano, Italy.*

Academic Service

- 2026  **Program chair.** STAG
Smart Tools and Applications in Graphics. *Catania, Italy*
-  **Member of program committee.** SGP
Symposium on Geometry Processing. *Bern, Switzerland*
-  **Member of program committee.** AICCSA
ACS/IEEE International Conference on Computer Systems and Applications. *Sharjah, UAE*
- 2025  **Poster chair.** STAG
Smart Tools and Applications in Graphics. *Genova, Italy*
-  **Member of program committee.** AICCSA
ACS/IEEE International Conference on Computer Systems and Applications. *Doha, Qatar*
-  **Member of program committee.** ECAI
European Conference on Artificial Intelligence. *Bologna, Italy*
- 2023  **Member of program committee.** TAG-ML
ICML's workshop on Topology, Algebra, and Geometry in Machine Learning. *Honolulu, Hawaii*
- 2022  **Member of RCDL Conference Coffee Committee.** ACM SIGGRAPH RCDL
ACM SIGGRAPH Research Career Development Committee. *Vancouver, Canada*
- 2021  **Event chair.** STAG
Smart Tools and Applications in Graphics. *Rome, Italy*

Teaching

- 2026  **Innovative Methods, Gamification, and Digital Applications**
Excellence School (4 hours). *Pegaso University*
-  **Teaching Networking**
Teaching qualification training course (9 hours). *Pegaso University*
- 2025 – Present  **Algorithms and Data Structures**
Undergrad course (30 hours/year). *Pegaso University*

Teaching (continued)

- 2025
 - **Networking and Cybersecurity**
Undergrad course (30 hours/year). *Pegaso University*
 - **Web Technologies**
Undergrad course (30 hours/year). *Pegaso University*
 - **Functional Maps: a spectral approach to the alignment of embeddings**
Ph.D. course (12 hours). *Sapienza – University of Rome*
 - **Digital Skills for Teaching**
Teaching qualification training course (12 hours). *Pegaso University*
 - **Teaching Networking**
Teaching qualification training course (18 hours). *Pegaso University*
- 2024 – 2025
 - **Networking and Cybersecurity**
Undergrad course (60 hours/year). *Pegaso University*
 - **Computer Architecture**
Undergrad course (30 hours/year). *Pegaso University*
- 2021
 - **Introduction to Algorithms**
Undergrad course (30 hours). *Sapienza – University of Rome*

Invited, Conference, and Seminars Talks

- April 2026
 - **The Ubiquity of Voronoi Diagrams**
Central European Seminar on Computer Graphics, 2026. *Smolenice, Slovakia*
- Nov 2025
 - **UV Parametrization via Topological Disk Segmentation of Surfaces**
Smart Tools and Applications in Graphics, 2025. *Genoa, Italy*
- Jun 2025
 - **Volumetric Functional Maps**
Sapienza – University of Rome, hosted by *E. Rodolà. Rome, Italy.*
- Nov 2024
 - **Scalable geometry processing in computer graphics applications**
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*
 - **TACO: a benchmark for connectivity-invariance in shape correspondence**
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*
 - **Efficient Generation of Multimodal Fluid Simulation Data**
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*
- Dec 2023
 - **A physically-inspired approach to the simulation of plant wilting**
ACM SIGGRAPH Asia, 2023. *Sydney, Australia.*
- Oct 2022
 - **MoMaS: mold manifold simulation for real-time procedural texturing**
Pacific Graphics (PG), 2022. *Kyoto, Japan.*
- May 2022
 - **Strassen's algorithm in practice**
Sapienza – University of Rome, hosted by *R. Marin. Rome, Italy.*
- Aug 2021
 - **Efficiently parallelizable Strassen-based multiplication of a matrix by its transpose**
International Conference on Parallel Processing (ICPP), 2021. *Chicago, Illinois, USA.*
- May 2021
 - **Orthogonalized Fourier polynomials for signal approximation and transfer**
EUROGRAPHICS (EG), 2021. *Vienna, Austria.*

Honours & Awards

- 2026  **CVPR 2026 Outstanding Reviewer**
- 2025  **STAG Best Paper Award – Honourable Mention**
UV Parametrization via Topological Disk Segmentation of Surfaces
Smart Tools and Applications in Graphics.
- 2024  **Matteo Dellepiane Award for PhD Thesis in Computer Graphics – Honourable Mention**
The Italian Chapter of EuroGraphics (EG-Italy).

Research Grants and Funding

- 2022  **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 2**
Principal investigator for the project *Enhancing Procedural Computer Graphics in Multimedia Applications with Fast Geometry Processing Techniques*.
- 2021  **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 1**
Principal investigator for the project *Automation of Casting Mold Design for Industrial Fabrication of Digital Objects*.
- 2020  **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 1**
Principal investigator for the project *GPU Fluid Simulation on Non-Euclidean Domains and Application for Simulation of Erosion Phenomena*.

Reviewing Service

- 2026  SIGGRAPH, NeurIPS, JOCCH, ECCV, JMIV, BMVC, SGP, TVCG, CVPR
- 2025  Pacific Graphics, ECAI, GCA, ICCV, CVPR, TPAMI, CAVW, EUROGRAPHICS
- 2024  ToG, ACCV, Pacific Graphics, ECCV, EUROGRAPHICS, CGF, TVCG
- 2023  Pacific Graphics, TAG-ML, ICCV, NeurReps, ICIAP
- 2022  EUROGRAPHICS

Supervision and Mentoring

- Ph.D.  Giulio Viganó (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Francesca Maccarone (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Daniele Baieri (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
Francesco De Canio (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
- M.Sc.  Giorgio Longari (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Roberta Giorgi (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
Thesis advisor of 5 students (Pegaso University)
- B.Sc.  Alessandro Sommariva (University of Milano-Bicocca) – Thesis co-advisor
Simone Pedico (University of Milano-Bicocca) – Thesis co-advisor
Alessio Tosato (University of Milano-Bicocca) – Thesis co-advisor
Arianna Silvestri (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Alireza Alipanah (Sharif University of Technology) – Internal supervisor (not formal advisor)
Daniele Solombrino (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
Thesis advisor of 11 students (Pegaso University)

Skills

Languages	Italian (mother tongue), English (professional proficiency).
Interpersonal	Adaptability to work independently and with(in) a team. Capability of supervising and communicating efficaciously. Excellent organizational and teaching abilities.
Programming	Proficient in C/C++ and MATLAB. Advanced knowledge of GPU programming with CUDA, GLSL, and HLSL. Knowledge of Python and C#.
Tools	Expert with the mesh processing software <i>MeshLab</i> and the rendering engine <i>Blender</i> . Advanced knowledge of the game engines <i>Unreal Engine 4 and 5</i> and <i>Unity 3D</i> . Familiar with software for raster (<i>GIMP</i>) and vector (<i>InkScape</i>) 2D graphics.

Research Publications

Journal Articles

- 1 D. Marin, **F. Maggioli**, S. Melzi, S. Ohrhallinger, and M. Wimmer, “Reconstructing curves from sparse samples on riemannian manifolds,” *Computer Graphics Forum*, vol. 43, no. 5, e15136, 2024.
- 2 **F. Maggioli**, R. Marin, S. Melzi, and E. Rodolà, “Momas: Mold manifold simulation for real-time procedural texturing,” *Computer Graphics Forum*, vol. 41, no. 7, pp. 519–527, 2022.
- 3 L. Moschella, S. Melzi, L. Cosmo, **F. Maggioli**, O. Litany, M. Ovsjanikov, L. Guibas, and E. Rodolà, “Learning spectral unions of partial deformable 3d shapes,” *Computer Graphics Forum*, vol. 41, no. 2, pp. 407–417, 2022.
- 4 **F. Maggioli**, S. Melzi, M. Ovsjanikov, M. M. Bronstein, and E. Rodolà, “Orthogonalized fourier polynomials for signal approximation and transfer,” *Computer Graphics Forum*, vol. 40, no. 2, pp. 435–447, 2021.
- 5 **F. Maggioli**, T. Mancini, and E. Tronci, “Sbml2modelica: Integrating biochemical models within open-standard simulation ecosystems,” *Bioinformatics*, vol. 36, no. 7, pp. 2165–2172, 2020.

Conference Proceedings

- 1 **F. Maggioli**, M. Livesu, and S. Melzi, “Volumetric functional maps,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- 2 D. Baieri, **F. Maggioli**, E. Rodolà, S. Melzi, and Z. Löhner, “Implicit-arap: Efficient handle-guided neural field deformation via local patch meshing,” in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- 3 A. Ferraris, F. Leveni, D. Baieri, **F. Maggioli**, S. Melzi, and L. Magri, “Geometric aware local optimization for robust primitive fitting,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2025.
- 4 **F. Maggioli**, D. Baieri, E. Rodolà, and S. Melzi, “Rematching: Low-resolution representations for scalable shape correspondence,” in *Computer Vision – ECCV 2024*, Cham: Springer Nature Switzerland, 2025, pp. 183–200, ISBN: 978-3-031-72913-3.
- 5 **F. Maggioli** and S. Melzi, “Uv parametrization via topological disk segmentation of surfaces,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2025.
- 6 D. Baieri, D. Crisostomi, S. Esposito, **F. Maggioli**, and E. Rodolà, “Efficient generation of multimodal fluid simulation data,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
- 7 F. Maccarone, G. Longari, G. Viganò, D. Peruzzo, **F. Maggioli**, and S. Melzi, “S4a: Scalable spectral statistical shape analysis,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.

- 8 S. Pedico, S. Melzi, and **F. Maggioli**, “Taco: A benchmark for connectivity-invariance in shape correspondence,” in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
- 9 **F. Maggioli**, J. Klein, T. Hädrich, E. Rodolà, W. Pałubicki, S. Pirk, and D. L. Michels, “A physically-inspired approach to the simulation of plant wilting,” in *SIGGRAPH Asia 2023 Conference Papers*, 2023, pp. 1–8.
- 10 **F. Maggioli**, D. Baieri, S. Melzi, and E. Rodolà, “Newton’s fractals on surfaces via bicomplex algebra,” in *ACM SIGGRAPH 2022 Posters*, 2022, pp. 1–2.
- 11 V. Arrigoni, **F. Maggioli**, A. Massini, and E. Rodolà, “Efficiently parallelizable strassen-based multiplication of a matrix by its transpose,” in *Proceedings of the 50th International Conference on Parallel Processing*, 2021, pp. 1–12.

Pre-prints

- 1 L. Olearo, G. Viganò, D. Baieri, **F. Maggioli**, and S. Melzi, “Fuse: A flow-based mapping between shapes,” arXiv preprint arXiv:2511.13431, 2025.
- 2 **F. Maggioli**, D. Baieri, Z. Lähner, and S. Melzi, “Sshade: A framework for scalable shape deformation via local representations,” arXiv preprint arXiv:2409.17961, 2024.
- 3 D. Baieri, S. Esposito, **F. Maggioli**, and E. Rodolà, “Fluid dynamics network: Topology-agnostic 4d reconstruction via fluid dynamics priors,” arXiv preprint arXiv:2303.09871, 2023.